:- DATA SCIENCE :-

Date
Page No.

(#) LLM :-

- chatgpt
- Gemini
- claude

(#) Big-data :-

Billion of users

(1) Programming language :-

- python, R

(2) Data processing :-

- Numpy, pandas

) Math & statistics :-

- By Tech classes
- Statistics
- probability
- Linear Algebra
- calculus

Data Visualization :-

- matplotlib / powerBI
- seaborn / tableau

Machine Learning :-

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning.

Scanned with OKEN Scanner

Reinforcement learning:

(when we trained algorithm with the help of Rewards, penalties)
ie cat, biscuit

supervised learning:

output defined (Patient is healthy or not) ← labeled data

→ Regression
Classification

Unsupervised learning:

unlabeled data
(there are no specific output is)
ie: (Health care sector employer does not able to explain patient is Healthy or not)

→ clustering (grp)
scikit learn library

Scanned with OKEN Scanner

Date
Page No.

⑥ Databases :

- SQL (Advanced level)

⑦ Version control :

- Git
- Github (good knowledge)

⑧ Deep Learning :

- Neural Network

⊛ Neural Network :

(Networks of neurons) which try to mimic the "human brain".

→ Networks of neurons (Billions of network)

- ANN (Artificial Neural Network)

- FNN (Feedforward Neural Network)
- RNN (Recurrent Neural Network) (NLP)
- Convolutional Neural Network (CNN)

- CNN :

To extract information from images, videos

Scanned with OKEN Scanner

Date
Page No.

* RNN (Recurrent Neural Network) :
it extract the information from text, textual content

* we learn D.L By using :
  - PyTorch
  - TensorFlow

- personal Recommendation
  in pyTorch for Deep Learning

*
Computer Vision
NLP
Generative AI

* pro tips :

- PROJECTS (proof of work)
- For Deployment Flask / FastAPI
- Hugging Face / streamlit
- Domain speciality.
- core DSA (Time span complexity)

Scanned with OKEN Scanner

:pipeline:

- Data collection (SQL)
- Data cleaning | preprocessing (python)
- Exploratory Data analysis (EDA) (Trend, Revenue growth)
- Feature Engineering (create new characteristic)
- model selection / training
- model Evaluation
- Model ~~deploy~~ deployment (MLOps)

Data Analyst:

(3+)
- Data collection (SQL, Excel)
- Data cleaning / preprocessing (python / statistics)
- EDA (powerBI / Tableau)

Data Scientist:

3+
- model selection / training

Scanned with OKEN Scanner

- model evaluation

(ii) AI Engineer:

primarily involve in neural network, large models

= Data scientist + software engineer

- PyTorch (✓)
- TensorFlow
- Keras

Scanned with OKEN Scanner